Servlet

1) Servlet Life Cycle using init(), service(), destroy()

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class LifeCycleServlet extends HttpServlet {

   public void init() throws ServletException {
      System.out.println("Servlet Initialized");
   }

   public void service(HttpServletRequest req, HttpServletResponse res)
         throws ServletException, IOException {

      res.setContentType("text/html");
      PrintWriter out = res.getWriter();

      out.println("<h2>Service Method Executed</h2>");
   }

   public void destroy() {
      System.out.println("Servlet Destroyed");
   }
}
```

---

2) Servlet using GenericServlet

```
import javax.servlet.*;
import java.io.*;

public class GenericServletExample extends GenericServlet {

   public void service(ServletRequest req, ServletResponse res)
         throws ServletException, IOException {

      res.setContentType("text/html");
      PrintWriter out = res.getWriter();

      out.println("<h2>Hello from Generic Servlet</h2>");
   }
}
```

---

3) Servlet using HttpServlet (GET and POST)

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class HttpServletExample extends HttpServlet {

   protected void doGet(HttpServletRequest req, HttpServletResponse res)
         throws ServletException, IOException {

      PrintWriter out = res.getWriter();
      out.println("<h3>This is GET request</h3>");
   }

   protected void doPost(HttpServletRequest req, HttpServletResponse res)
         throws ServletException, IOException {

      PrintWriter out = res.getWriter();
      out.println("<h3>This is POST request</h3>");
   }
}
```

---

4) RequestDispatcher using forward()

First Servlet

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class ForwardServlet extends HttpServlet {

   protected void doGet(HttpServletRequest req, HttpServletResponse res)
         throws ServletException, IOException {

      RequestDispatcher rd = req.getRequestDispatcher("SecondServlet");
      rd.forward(req,res);
   }
}
```

Second Servlet

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class SecondServlet extends HttpServlet {

   protected void doGet(HttpServletRequest req, HttpServletResponse res)
```

```
        throws ServletException, IOException {

    PrintWriter out = res.getWriter();
    out.println("<h2>Request forwarded successfully</h2>");
  }
}
```

---

5) RequestDispatcher using include()

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class IncludeServlet extends HttpServlet {

  protected void doGet(HttpServletRequest req, HttpServletResponse res)
        throws ServletException, IOException {

    PrintWriter out = res.getWriter();

    out.println("<h3>Main Servlet Content</h3>");

    RequestDispatcher rd = req.getRequestDispatcher("SecondServlet");

    rd.include(req,res);

    out.println("<h3>After Include</h3>");
  }
}
```

---

6) Session Tracking using Cookies

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class CookieExample extends HttpServlet {

  protected void doGet(HttpServletRequest req, HttpServletResponse res)
        throws ServletException, IOException {

    Cookie ck = new Cookie("username","Pranjali");

    res.addCookie(ck);

    PrintWriter out = res.getWriter();
```

```
        out.println("Cookie added successfully");
    }
}
```

---

7) Session Tracking using Hidden Form Fields

HTML

```
<form action="HiddenServlet">
<input type="hidden" name="username" value="Pranjali">
<input type="submit" value="Submit">
</form>
```

Servlet

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class HiddenServlet extends HttpServlet {

    protected void doGet(HttpServletRequest req, HttpServletResponse res)
            throws ServletException, IOException {

        String name = req.getParameter("username");

        PrintWriter out = res.getWriter();
        out.println("User Name: " + name);
    }
}
```

---

8) URL Rewriting Example

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class URLRewriteServlet extends HttpServlet {

    protected void doGet(HttpServletRequest req, HttpServletResponse res)
            throws ServletException, IOException {

        PrintWriter out = res.getWriter();

        String name = req.getParameter("name");

        out.println("Welcome " + name);
```

```
        out.println("<a href='NextServlet?name="+name+"'>Next Page</a>");
    }
}
```

---

9) Session Tracking using HttpSession

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class SessionExample extends HttpServlet {

    protected void doGet(HttpServletRequest req, HttpServletResponse res)
            throws ServletException, IOException {

        HttpSession session = req.getSession();

        session.setAttribute("user","Pranjali");

        PrintWriter out = res.getWriter();
        out.println("Session stored successfully");
    }
}
```

---

10) Session Management with HttpSession

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class SessionDemo extends HttpServlet {

    protected void doGet(HttpServletRequest req, HttpServletResponse res)
            throws ServletException, IOException {

        HttpSession session = req.getSession();

        session.setAttribute("name","Pranjali");

        String name = (String) session.getAttribute("name");

        PrintWriter out = res.getWriter();
        out.println("User Name: " + name);
    }
}
```

Advantages of HttpSession

- Maintains user session automatically
- No need to send data repeatedly
- More secure than cookies
- Stores objects on the server side

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13) Servlet with JDBC to display records

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;
import java.sql.*;

public class DisplayDataServlet extends HttpServlet {

   protected void doGet(HttpServletRequest req, HttpServletResponse res)
         throws ServletException, IOException {

      PrintWriter out = res.getWriter();

      try {

         Connection con = DriverManager.getConnection(
               "jdbc:mysql://localhost:3306/testdb","root","");

         Statement stmt = con.createStatement();

         ResultSet rs = stmt.executeQuery("SELECT * FROM student");

         while(rs.next()) {

            out.println(rs.getInt(1)+" "
                  +rs.getString(2)+" "
                  +rs.getInt(3)+"<br>");
         }

         con.close();
      }
      catch(Exception e) {
         out.println(e);
      }
   }
}
```

---

14) Insert form data using PreparedStatement

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;
import java.sql.*;

public class InsertServlet extends HttpServlet {

   protected void doPost(HttpServletRequest req, HttpServletResponse res)
         throws ServletException, IOException {

      String name = req.getParameter("name");
      String age = req.getParameter("age");

      PrintWriter out = res.getWriter();

      try {

         Connection con = DriverManager.getConnection(
               "jdbc:mysql://localhost:3306/testdb","root","");

         PreparedStatement ps = con.prepareStatement(
               "INSERT INTO student(name,age) VALUES(?,?)");

         ps.setString(1,name);
         ps.setString(2,age);

         ps.executeUpdate();

         out.println("Record Inserted");

         con.close();
      }
      catch(Exception e) {
         out.println(e);
      }
   }
}
```

---

15) Display database records in HTML table

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;
import java.sql.*;

public class TableServlet extends HttpServlet {
```

```
    protected void doGet(HttpServletRequest req, HttpServletResponse res)
        throws ServletException, IOException {

      PrintWriter out = res.getWriter();

      try {

        Connection con = DriverManager.getConnection(
              "jdbc:mysql://localhost:3306/testdb","root","");

        Statement stmt = con.createStatement();

        ResultSet rs = stmt.executeQuery("SELECT * FROM student");

        out.println("<table border=1>");

        while(rs.next()) {

          out.println("<tr>");
          out.println("<td>"+rs.getInt(1)+"</td>");
          out.println("<td>"+rs.getString(2)+"</td>");
          out.println("<td>"+rs.getInt(3)+"</td>");
          out.println("</tr>");
        }

        out.println("</table>");

        con.close();
      }
      catch(Exception e) {
        out.println(e);
      }
    }
}
```

---

16) Update database records using Servlet

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;
import java.sql.*;

public class UpdateServlet extends HttpServlet {

    protected void doPost(HttpServletRequest req, HttpServletResponse res)
        throws ServletException, IOException {
```

```
        int id = Integer.parseInt(req.getParameter("id"));
        String name = req.getParameter("name");

        PrintWriter out = res.getWriter();

        try {

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/testdb","root","");

            PreparedStatement ps = con.prepareStatement(
                "UPDATE student SET name=? WHERE id=?");

            ps.setString(1,name);
            ps.setInt(2,id);

            ps.executeUpdate();

            out.println("Record Updated");

            con.close();
        }
        catch(Exception e) {
            out.println(e);
        }
    }
}
```

---

17) How Servlets interact with Database using JDBC

Explanation

Servlets can interact with databases using **JDBC (Java Database Connectivity)**.

Steps

1. Load JDBC driver
2. Establish database connection
3. Create Statement or PreparedStatement
4. Execute SQL query
5. Process ResultSet
6. Close connection

Advantages of Servlets over CGI

Servlet	CGI
Faster execution	Slower
Runs inside server	Creates new process each request
Better memory usage	High resource usage
Supports multithreading	No multithreading
Easy database connectivity	More complex